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INSO4998 – Undergraduate Research in
Software Engineering
Quantum Cryptography

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Goal: After the last meeting, we were tasked with researching the following metrics: Layer Fidelity (or Layered Gate Error), coherence times (T_1 , and T_2), connectivity (topology), T-count and T-depth, logical error rate.

Actions:

Juan D. Guadalupe Rosado

Present:

- I have begun researching information regarding the mentioned metrics.

Future:

- Implement the metrics so that we can measure the performance of circuits that we have previously made/measured. I hope to have a demo ready for the next meeting.

Daniel C. Colón Rivera

Present:

- Re-read the paper: [A Comprehensive Study of Quantum Arithmetic Circuits](#).
- Selected circuits to implement: Wang's 2024 Carry-Lookahead design for addition, subtraction will be a direct modification of addition circuit, one of Nie's 2023 designs for multiplication, and one of Thapliyal's 2019 designs for division. (Tentative)
- Implemented Wang's 2024 optimal addition circuit and successfully simulated Statevector extraction. Seems to be correct.

Future:

- Finalize decision of and implement all selected circuits.
- Create functions to extract results for each statevector.
- Run circuits on IBM quantum computers.